

1. Final Training Results

Our final results show ~85% accuracy when using a binary classifier which predicts whether a comment was posted in between the years 2008-10 or 2020-2022. This is better than the preliminary results, which showed 72% accuracy when guessing between 2013 and 2024. To make this improvement happen, we changed the pretrained weights from distilBERT to roBERTa-base. RoBERTa-base is a better model for our case because it is trained on a wider variety and significantly more data than distilBERT. Additionally, we added compatibility with cuda GPUs for faster training on Kaggle Notebooks with T4 GPUs.

Here are the final results, accuracy: 85.3%, precision: 87.3%, F1: 85.5%.

2. Final demonstration proposal

For our final demonstration, we built a full-stack web application that showcases our Reddit Year Prediction model. The site includes an interactive demo where users can input Reddit-style comments and receive real-time predictions indicating whether the text is from 2008–2010 or 2020–2022, along with confidence scores. The frontend is built with Next.js, TypeScript, Tailwind, and Shadcn UI, giving us a modern, responsive interface, while the backend uses FastAPI with PyTorch and Hugging Face Transformers to serve our fine-tuned RoBERTa 2-bin classifier (85% accuracy), with a regression fallback model for robustness. User input flows from the frontend to a /predict API route, which proxies the request to the FastAPI backend where the model performs inference and returns probabilities that are immediately displayed to the user. We also implemented model caching, CORS handling, and an ngrok tunnel to allow remote demonstrations. Through this project, we learned how to integrate an ML model into a production-style web stack, manage API communication, and design an intuitive interface that effectively showcases our model's capabilities.